

MENRA WEDWANG ROMIAL

Published as *Romial Menra*

PhD Candidate in Computer Science, Energy-Aware Cloud Computing

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RESEARCH PROFILE

PhD candidate working on power management in cloud infrastructures, from hardware-level power limiting (Intel RAPL, Turbo Boost) to energy-aware orchestration in Kubernetes. My work combines large-scale experimentation on Grid'5000, analytical modelling of processor power behaviour, and the design of open-source tools that make power a first-class resource in the cloud. Before the PhD, two years as a full-stack developer and team lead gave me a practical, production-oriented view of the systems I now study.

Keywords: power capping, Intel RAPL, energy efficiency, Kubernetes, scheduling, distributed systems, green IT.

EDUCATION

PhD in Computer Science

Nov. 2024 – present

IMT Atlantique, Inria, LS2N, Nantes, France

Thesis: *Contribution à la gestion de la puissance dans les infrastructures cloud : de la limitation matérielle à l'orchestration adaptative.*

Master of Engineering in Computer Science

2021 – 2024

National Advanced School of Engineering of Yaoundé (ENSPY), Cameroon

Specialisation in Distributed Systems and Software Engineering.

Preparatory Classes, Mathematics & Physics

2019 – 2021

ENSPY, Cameroon

PUBLICATIONS

International conferences

[C2] **Romial Menra**, Guillaume Rosinosky, Remous-Aris Koutsiamanis, Sébastien Bolle, Jean-Marc Menaud. **Understanding Power Limiting Mechanisms in Modern Processors: A Deep Dive Into Intel RAPL and Turbo Boost Dynamics.** *Euro-Par 2026: Parallel Processing*, Lecture Notes in Computer Science, Springer, pp. 135–149, 2026. **Best Paper Award nominee.** Acceptance rate 26.2%.
[doi:10.1007/978-3-032-35251-4_10](https://doi.org/10.1007/978-3-032-35251-4_10) · [hal-05729597](https://hal.archives-ouvertes.fr/hal-05729597)

National conferences

[N1] **Romial Menra**, Remous-Aris Koutsiamanis, Jean-Marc Menaud. **Intégration de l'Aspect Énergétique dans Kubernetes.** *COMPAS 2024, Conférence francophone d'informatique en Parallélisme, Architecture et Système*, Nantes, France, 2024.
[hal-05638939](https://hal.archives-ouvertes.fr/hal-05638939)

TALKS

Understanding Power Limiting Mechanisms in Modern Processors

Aug. 2026

Euro-Par 2026, 32nd International European Conference on Parallel and Distributed Computing, Pisa, Italy

Energy-Aware Kubernetes Orchestration for Grid-Responsive Computing

Oct. 2025

STACK team seminar, Arzon, France

Intégration de l'Aspect Énergétique dans Kubernetes

Jul. 2024

COMPAS 2024, Nantes, France

PROFESSIONAL EXPERIENCE

PhD Candidate

Nov. 2024 – present

Inria, IMT Atlantique, LS2N, Nantes, France

- Characterised the interaction between Intel RAPL power limits and Turbo Boost on four Intel Xeon microarchitectures, and derived an analytical model of turbo duration.
- Showed that power capping below about 50% of TDP systematically degrades energy efficiency.

- Designing Kubernetes components for power-aware operation: node-level power capping, automatic server power profiling and energy-aware scheduling.
- Large-scale experimentation on the Grid'5000 testbed; collaboration with Orange Research.

Research Intern

Apr. 2024 – Aug. 2024

IMT Atlantique, Nantes, France

- Implemented server power limitation with Intel RAPL and evaluated it on Grid'5000.
- Developed an energy-aware Kubernetes scheduler with custom resource constraints.
- Results published at COMPAS 2024.

Software Developer & Team Lead

May 2022 – Mar. 2024

Forall Founders, Cameroon

- Led a team of 10 developers building e-learning platforms and school management systems (Django, Node.js, React, Next.js, Vue.js).
- Project manager of a mobile e-learning application with 15,000+ active users (**TEFAA**); cut latency from 5 s to 1.2 s through query optimisation and caching.
- Built an IoT attendance system (PyQt, Raspberry Pi, Arduino) and CI/CD pipelines on Linode.

TEACHING & SUPERVISION

Teaching Assistant, Databases (TD/TP)

Spring 2025

IMT Atlantique, Nantes, first-year engineering students

Relational model, SQL, normalisation, UML conceptual modelling, programmatic interfaces.

Teaching Assistant, Cloud Computing with VMware vSphere (TP)

Spring 2025

IMT Atlantique, Nantes

Co-supervision, Clement Obama (M2 research intern)

Apr. 2026 – Sep. 2026

IMT Atlantique, Nantes

Reactive control of energy constraints in Kubernetes clusters: multi-lever re-adaptation policy with SLA management.

SERVICE & COMMUNITY

- Student volunteer, UCC & BDCAT 2025 (Utility and Cloud Computing / Big Data Computing, Applications and Technologies), Nantes, Dec. 2025.
- Founder of **C5IN**, Cameroon Cloud-Edge-IoT Innovation Network.
- Contributor to **DIBAC**, an open-access institutional repository for Cameroonian research, and **TupuriHub**, a digital hub for Toupouri culture and heritage.
- Technical blog on Kubernetes and Linux power management at menraromial.com.

TECHNICAL SKILLS

Research & HPC

Grid'5000, Intel RAPL, powercap, energy profiling, performance benchmarking

Cloud & DevOps

Kubernetes (k8s, k3s, kubeadm, Kubebuilder, scheduler plugins), Docker, Terraform, Ansible, GitHub Actions, GitLab CI, Jenkins, Prometheus, Grafana

Languages

Go, Python, C, Java, JavaScript/TypeScript, Bash

Web

Django, FastAPI, Node.js, React, Next.js, Vue.js, Spring Boot

Data

PostgreSQL, MySQL, MongoDB, Redis, Pandas, NumPy

CERTIFICATIONS & INTERESTS

Certifications: IBM (Coursera, 2023): Agile Development and Scrum, Applied Data Science Capstone, Machine Learning with Python, Data Visualization with Python, Databases and SQL for Data Science. EPFL (Coursera, 2021): Programming and Object-Oriented Programming in Java.

Interests: open source, technical writing, reading (science, leadership, history), documentaries, cooking, cinema, nature.